

Monday 28th October 2025

On **Monday 27th October 2025** I had my first site experience with a local electrician in Leicester. The scope of the project was a domestic setting. The end user has plans to transform an outhouse into a social space with a cocktail bar and lounge area. On the day the first task I had to do was drill a seventy millimetre hole in the ceiling. It was a total of twelve holes. Above the ceiling lay a doncaster earth sure cable that I had to pull through the ceiling hole. An "earthsure cable" differs slightly from the twin and earth cable that I used in college. The main difference is that the yellow and green wire has insulation already on the wire. Whereas, the twin and earth I use in college requires earth sleeving to be applied to the bronze copper wire.

After I pulled the cable through the ceiling I had to strip the cable and expose the copper to then screw into the terminals. There were two sets of wires in each terminal port apart from the final bulb that only had one set of cpc, blue and brown wire. Because the lighting is on a radial circuit. Which means that the cable does not return back to the fuse board. You know are at the end of the radial circuit when there is only one set of wires.

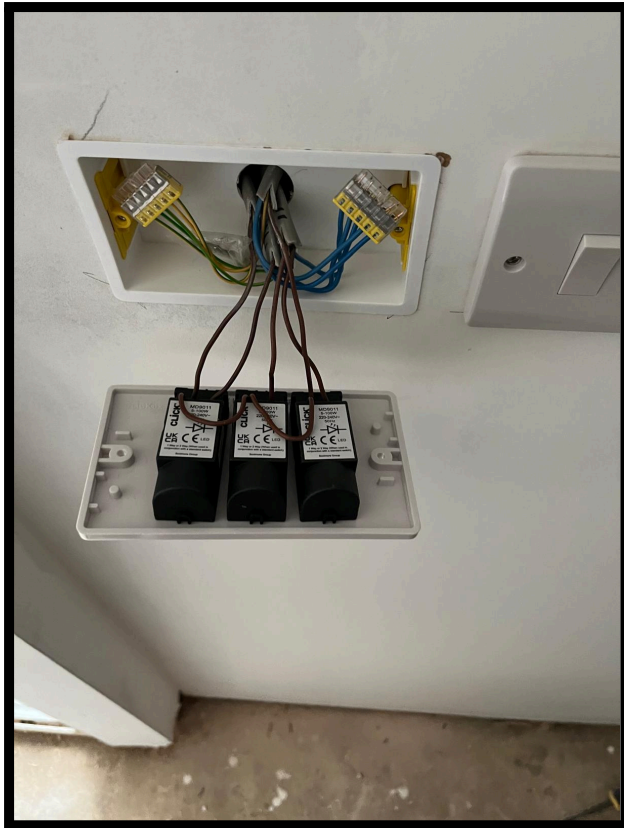
The cpc insulation is slightly smaller than the brown live and blue neutral wire because the earth is only designed to carry current in the event of a fault.

I also wired up two outdoor lights. The outer-house has a sub-panel that was installed which fed off the main consumer unit for the household. We used the SWA armour cable which connected the main fuse board (which is in the main dwelling) to the sub-fuse board (located inside the outer house).

The voltage delivered to houses across the UK is 230 volts. However, oftentimes the voltage is closer to 240 volts.

In conclusion this was a great electrical site experience. Because, thank you for the local electrician for the opportunity and for the tasks that I did that day.

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